

Task 1

We want to solve the regularized least-squares problem

$$\min_x f(x) := \|Ax - b\|_2^2 + \lambda^2 \|x\|_2^2.$$

Expanding the objective gives

$$f(x) = (Ax - b)^T(Ax - b) + \lambda^2 x^T x = x^T A^T A x - 2x^T A^T b + b^T b + \lambda^2 x^T x.$$

Since f is quadratic (and convex), $\nabla f(x) = 0$ is a necessary and sufficient condition for optimality:

$$\nabla f(x) = 2A^T A x - 2A^T b + 2\lambda^2 x = 0.$$

This simplifies to the normal equations

$$(A^T A + \lambda^2 I)x = A^T b.$$

Let $A = UEV^T$ be the SVD of A . Substituting into the normal equations and using $U^T U = I$:

$$\begin{aligned}(A^T A + \lambda^2 I)x_{\text{RLS}} &= A^T b \\ (VE^T U^T UEV^T + \lambda^2 I)x_{\text{RLS}} &= VE^T U^T b \\ V(E^2 + \lambda^2 I)V^T x_{\text{RLS}} &= VE^T U^T b \\ (E^2 + \lambda^2 I)x_{\text{RLS}} &= E^T U^T b\end{aligned}$$

Let $B := E^2 + \lambda^2 I$, which is diagonal. Then

$$x_{\text{RLS}} = B^{-1} E^T U^T b.$$

Since B is diagonal, this holds component-wise: writing $y_i := v_i^T x_{\text{RLS}}$,

$$y_i = \frac{\sigma_i}{\sigma_i^2 + \lambda^2} u_i^T b.$$

This gives us the final relation

$$x_{\text{RLS}} = \sum_{i=1}^n v_i \frac{\sigma_i}{\sigma_i^2 + \lambda^2} u_i^T b = Vy.$$

The filter factor $\frac{\sigma_i}{\sigma_i^2 + \lambda^2}$ can be interpreted as a low-pass filter: as $\lambda \ll \sigma_i$ it approaches 1, at $\lambda = \sigma_i$ it equals $\frac{1}{2}$, and as $\lambda \gg \sigma_i$ it approaches 0.

Task 2

Starting from the (unregularized) least-squares solution expressed via the SVD,

$$x_{\text{LS}} = \sum_{i=1}^n \frac{u_i^T b}{\sigma_i} v_i,$$

small singular values σ_i can amplify noise in b . As in Task 1, we introduce a filter factor $\varphi_i(\lambda)$ that damps the contribution of each term:

$$y_i = \varphi_i \frac{\lambda}{\sigma_i} u_i^T b, \quad y = \sum_{i=1}^n y_i v_i = \sum_{i=1}^n \varphi_i \frac{\lambda}{\sigma_i} (u_i^T b) v_i.$$

For the truncated SVD, the filter factor is chosen as

$$\varphi_i(\lambda) = \begin{cases} 1 & \text{if } i \leq k \\ 0 & \text{if } i > k. \end{cases}$$

Substituting this in, only the first k terms survive:

$$Vy = \sum_{i=1}^k \frac{u_i^T b}{\sigma_i} v_i + \sum_{i=k+1}^n 0 \cdot \frac{u_i^T b}{\sigma_i} v_i = x_{\text{TLS}}.$$